Q6 **NO CALCULATOR ALLOWED** Q6

Answer QUESTION 6 PARTS A and B on this page.

PART A

$$\lim_{n\to\infty}\left|\frac{\dfrac{(x-4)^{n+2}}{(n+2)\,3^{n+1}}}{\dfrac{(x-4)^{n+1}}{(n+1)\,3^{n}}}\right| = \lim_{n\to\infty}\left|\frac{(x-4)\times(n+1)}{(n-2)\;3}\right| = |x-4|\lim_{n\to\infty}\left|\frac{n+1}{3(n-2)}\right|$$

due to ratio test.

$|x-4|\cdot\frac{1}{3}<1$, $-3<x-4<3$, $\boxed{1<x<7}$

PART B

$$f' \approx \frac{2(x-4)}{2\cdot 3} + \frac{3(x-4)^2}{3\cdot 3^2} + \frac{4(x-4)^3}{4\cdot 3^3} + \cdots + \frac{(n+1)(x-4)^n}{(n+1)\,3^n} + \cdots$$

$$\boxed{f' \approx \frac{(x-4)}{3} + \frac{(x-4)^2}{3^2} + \frac{(x-4)^3}{3^3} + \cdots + \frac{(x-4)^n}{3^n} + \cdots}$$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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Answer QUESTION 6 PARTS C and D on this page.

PART C

Since f' is Geometric series $\left(S=\frac{a}{1-r}\right)$. $\left(\left|\frac{x-4}{3}\right|<1\right.$ at H converges to $\left.\right)$

$a=\frac{(x-4)}{3}$ $r=\frac{x-4}{3}$.

$$S=\frac{\left(\frac{(x-4)}{3}\right)}{\left(\frac{3-x+4}{3}\right)}=\frac{x-4}{7-x}$$

$$\frac{\frac{x-4}{3}}{1-\left(\frac{x-4}{3}\right)}$$

$\therefore f'(x)=\frac{x-4}{7-x}$

PART D

$f'(x)=\frac{x-4}{7-x}$ @ $x=8$?

No. Since $f'(x)=\frac{x-4}{7-x}$ is derived from where $x=4$ (center is 4) the $T(x)$ general form is $\frac{(x-4)^n}{3^n}$. However, since the center is shifted to $x=8$, In $T(x)$, $(x-8)$ is needed to be used instead of $(x-4)$. Leading to different convergence $f'(x)$ formula.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.